

Weekly Self-Study Tasks to Complete

Phase 2 — Week 2: TypeScript, NPM, and Vite for Leaders (Part 1 of 2): TypeScript and NPM

This document covers Part 1 (TypeScript and NPM) of Week 2 of Phase 2 in the 4-hour Leaders model: 2 hours of instructor-led session, 1 hour of self-study video, and 1 hour of hands-on lab. The Core Resources table below fits inside the 1-hour self-study media budget. The Practice Activities section points to the Evergreen Insurance Quote Page lab, where you will add types to the quote app and earn a green build. Additional Content at the end is optional and sits outside the 4-hour budget. This TypeScript, NPM, and Vite course spans two weeks; Part 2 goes deeper on Vite, build tooling, and project architecture in the companion Vite and Build Tooling Weekly Tasks document.

Core Resources (~60 min)

Watch the specific resources called out below — not an entire course. Keep VS Code open as you watch so the red type-error squiggles the videos describe are real to you.

Topic	Material Description	Est. Time	Link / Reference
TypeScript Big Picture	TypeScript: The Big Picture (Simon Allardice, Pluralsight, updated Jan 12, 2026; 44m course). Watch “How TypeScript Can Help” (~14 min) and “Writing TypeScript Applications” (~15 min). A concept-first overview of what TypeScript is, why teams adopt it, and how it compiles to JavaScript — no hands-on required.	~29 min	pluralsight.com — TypeScript: The Big Picture
Types & Interfaces	The TypeScript Handbook — “The Basics” and “Everyday Types” (official docs). Read and skim how basic	~16 min	typescriptlang.org — Handbook

Topic	Material Description	Est. Time	Link / Reference
	types, interfaces, and unions add a safety net to JavaScript. Free and authoritative.		
NPM & Vite	The Vite “Getting Started” guide (official) — read and skim the npm scripts (npm install, npm run dev, npm run build) that power a modern TypeScript project. Free and authoritative.	~15 min	vite.dev — Getting Started

Practice Activities (1 hour)

Complete this week’s hands-on lab: “Evergreen Insurance Quote Page — Adding Types with TypeScript, NPM, and Vite.” You will install the project with npm, run it with Vite, and add the types that make the build pass. The whole exercise takes about one hour.

In the lab you will:

- Open the Labfiles/evergreen-quote-typescript-build folder and run npm install, then npm run dev.
- See the red type errors in the editor and the failing npm run build.
- Define a CoverageType union and a Quote interface in src/types.ts.
- Add type annotations to calculatePremium and validateInputs in src/premium.ts.
- Pass a string where a number belongs and watch TypeScript catch it in the editor — before running.
- Undo the bug and earn a green npm run build, then npm run preview the result.

The lab instructions (with “Why this matters” callouts throughout) are in the lab environment. Budget one hour and work at your own pace.

Additional Content (Optional)

The following resources are optional and recommended for students who want to go deeper beyond the core curriculum. They are outside the 4-hour weekly budget. React (taught with TypeScript) is covered in the next weeks of Phase 2, so focus this week on types, interfaces, and the npm/Vite workflow.

Topic	Material Description	Link / Reference
Reference	The TypeScript Handbook — the official, authoritative guide to the language, from basics to advanced types (Reference / Tutorial).	typescriptlang.org/docs/handbook
Try It Live	The TypeScript Playground — write TypeScript in the browser and watch the compiled JavaScript and type errors update live (Tool).	typescriptlang.org/play
NPM	An Introduction to the npm package manager — Node.js official docs. What npm is, package.json, and the install/run workflow (Tutorial).	nodejs.org — Introduction to npm
Vite	Vite Guide — the official “Getting Started” and “Why Vite” pages covering the dev server and the production build (Reference).	vite.dev/guide